A Low-noise High-gain Transimpedance Amplifier With High Dynamic Range in 0.13ìm CMOS

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Abstract — This paper designed a low-noise high-gain transimpedance amplifier with high dynamic range. The input stage of TIA uses an individual power supply to accommodate the level of the photodiode(PD). Then the signal is transmitted to the subsequent circuit by a level shifting circuit. The TIA adopts DCrestore to stabilize work points, and uses a MOS working in the linear region to reduce the effective input signal to realize high dynamic range. Key noise components are optimized to reduce the input referred current noise. This TIA was validated in 0.13 um CMOS. The simulation results show the -3dB bandwidth is 1.8GHz, the maximum gain is 82.27dBΩ, the input referred noise is 125nA, and the measure current ranges 5uA ~2mA.

Index Terms — Complementary metal - oxide - semiconductor(CMOS), transimpedance amplifier(TIA), low-noise, high-gain, dynamic range

I. INTRODUCTION

In optical fiber communication system, the preamplifier is a key part in optical receiver, which detects the weak current signal and transforms it to voltage signal with low noise. Because of the different transmitting power and the different of fiber length, the power of received signal may differ greatly. So the TIA should own a wide input range[1]. Besides, to detect weak signal from long fiber, the equivalent input noise should be low enough[2][3].

TIA topologies includes the passive TIA and the active TIA[4].A passive TIA is a single resistance which is in parallel with the photodiode. The resistance R and the photodiode capacitance C_{PD} decide the -3dB bandwidth ($BW_{-3dB}=1/(RC_{PD})$). The passive TIA can directly convert the photocurrent to logic levels with zero power consumption[4]. However, the resistance should be large enough for weak photodiode current. So the bandwidth is limited severely.

The traditional passive TIA topologies include a common gate (CG) TIA[5], a common source (CS) TIA[6][7], a common drain (CD) TIA[8], a common gate TIA with active feedback (CGAF, typical example: RGC TIA)[9], ect. The paper[10] shows the -3dB bandwidth and the input referred current noise of the four topologies.

The four topologies were realized for Gigahertz TIA in many papers[5~10]. However, the DC current affects the TIA quiescent operating point.What's more, the simulation shows the variational DC current may introduce noise[11]. Because of wide bandwidth requirement in high speed signal transfer, it rarely use capacitive coupling method to get rid of the DC influence.

This paper adopts DCrestore to make the DC level of feed-back resistance almost equal, which can reduce the impact of changes in the PD DC current. In the meanwhile, the input stage uses a different power supply to accommodate the level of the PD, and uses a MOS working in the linear region to reduce the input signal in the input stage to realize high dynamic range. Then a level shifting circuit transmits the signal to a normal level for subsequent circuits. To reduce the equivalent input noise, key components are optimized properly.

The TIA in this paper is verified in 0.13ì m CMOS. Its area is about 0.012mm^2. The simulation results showed that the -3dB bandwidth is 1.8GHz, the maximum gain is 82.27dBÙ, the input referred current noise is 125nA in the range of 1Hz ~5GHz, and it can measure current from 5uA to 2mA. It can be used in optical fiber communication system, and it can work at data transfer rate of 2.5 Gbps.

II. PROPOSED TIA TOPOLOGY

This paper uses three stages to convert the photocurrent to voltage signal. The core circuit is as fig 1(in this paper, the substrate of NMOS without drawing the substrate port is linked with gnd; the substrate of PMOS without drawing the substrate port is linked with VDD2). The first stage is the input stage, which realizes gain control of the amplifier. The second stage is a level shifting circuit, and shifts the TIA quiescent operating point. The third stage is a source follower. The feed-back resistance Rf realizes transimpedance amplification.

In the fig 1, M0 and R0 of the first stage constitute the input DC bias circuit; M0 gate voltage can control the M0 current to adjust the bias voltage. M2 works in the linear region, so the equivalent resistance can be changed by its gate voltage; it can adjust the gain. In the second stage, M6 and R8 form a current source to stabilize M3 operating point; the small signal is shifted to VDD2 by R3;

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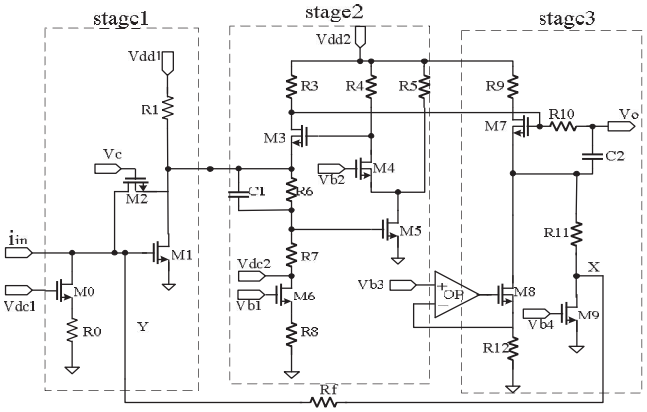


Fig. 1. The core circuit of proposed TIA

M4 and M5 feed back to M3 to stabilize the transconductance of M3; M6 reduces the voltage to bias the M5; R5 divides up part current of M5, so it stabilize the voltage level of the M5's drain. Capacitors C1 and R6 in parallel prevent oscillation in the closed-loop formed by M3、M4、M5. In the third stage, R10 makes the output DC voltage level equal to the output DC voltage level of the second, so the second stage decides the output DC voltage level of TIA; because R10 is so large that it can pass DC but suppress the small signal, the small signal is coupled to Vo by C2; M6、OP and R12 forming a stable current source, mainly decide the working point of M8 and reduce the current of R11. The DC voltages on the two terminals of Rf almost equal by the adjustment of DCrestore, so only small signal is fed back to input by Rf.

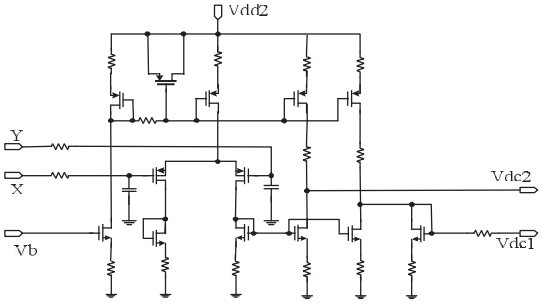
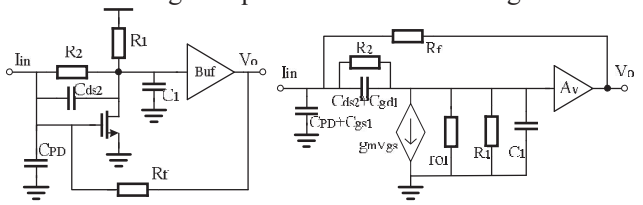


Fig. 2. DCrestore circuit

The working principle of DCrestore is to make the DC voltages on the two terminals of Rf almost equal through the circuit as fig 2 which detects the DC voltages on the two terminals of Rf, so only small signal is fed back to input by Rf, which makes TIA work normally. When the DC current of PD increases, the voltage leve of Y point become high, so DCrestore makes Vdc1 and Vdc2 increase; high Vdc1 lets M0 vent out some current to reduce the level of Y point, and high Vdc2 lets the output DC level of the second stage increase, so the level of point X increases too. This makes the DC voltages on the two terminals of Rf almost equal, only small signal pass Rf. And vice versa.

III. SMALL SIGNAL ANALYSIS

Because the gain of the level shifting circuit and the output stage is almost stable, and the output resistance is decided by the small resistance R1(only dozens ohms), so the second stage and the third stage can be equivalent to a buffer. M2 works in the linear region, so it can be replaced by a resistance R2. So fig1 can be equivalent to fig 3a, and the the small signal equivalent circuit is like fig 3b.



(a) (b)

Fig. 3. The equivalent circuits of TIA

According to Fig 3, the transconductance gain can be deduced as follow:

$$\frac{V_o}{i} \approx \frac{A_v R_1 \left(R_f // R_2 // \frac{1}{(C_{PD}+C_{gs1}+C_M)s} \right)}{R_2 // R_f // \frac{1}{C_{gd}s}} // \left[-A_v (R_2 // \frac{1}{C_M}) \right] // (-R_f) \quad (1)$$

Av is the gain of buffer; CM is the sum of the parasitic capacitance of M1, Cgd1 and the parasitic capacitance of M2, Cds2. The small signal equivalent resistance of MOS working in the saturation region ro is so large that it can be neglected when it is in parallel with other smaller resistances. R1 is less than other resistances much, so others in parallel with it can be neglected.

According to the gain equation, Rf and R2 mainly decide the gain. Rf is fixed, so the gain is controlled by R2. It can be considered as R2(M2) divides some photocurrent from PD, so the photocurrent passing Rf reduces, which affects the gain.

Because the capacity of PD is so large that it decides the bandwidth of TIA, the typical value of which is 0.1pF to 1 pF, the bandwidth expression is:

$$\omega_{-3db} = \frac{1}{R_{in} C_{in}} = \frac{1}{(R2 // Rf)(C_{PD} + C_{gs1} + C_M g_{m1} R1)} \quad (2)$$

Referencing literatures [12][13][14], the input referred noise can be expressed:

$$S_{eq}(f) = \frac{4kT}{R_1} + \frac{4kT}{R_f} + \frac{4kT}{R_2} + 2qI_{g1} + \frac{4kT\Gamma}{g_{m1}} \left[\frac{1}{R_1^2} + \frac{1}{R_f^2} + \frac{1}{R_2^2} + (2\pi f)^2 (C_{pd} + C_{gs1})^2 \right] \quad (3)$$

It can ignore the little term $2qg_1$, which represent the noise contributions due to gate leakage currents. The noise of buffer needs divided by the gain of the first stage, so it should affect little to the input referred noise. So M1、Rf and M2 are key to the input referred noise. According to $g_m = \mu C_{ox}(W/L)V_{ov}$, it can increase the W/L to reduce the noise. In this paper, the W/L of M1 is 864.

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Increasing Rf and R2(M2) can reduce the noise too, but they also increase the equivalent input resistance, which reduces the bandwidth. So it should trade off between the noise and the bandwidth.

IV. LAYOUT AND SIMULATION

The layout of TIA is shown in Fig 4. The size is approximately 0.012mm^2(excluding pads and ESD; the whole area is 232*150um^2). There uses thick isolation to prevent key devices affected by outside noise. The feedback resistance and R6 in the second stage are formed by part p type polysilicon resistor and part n type polysilicon resistor to temperature compensate, which reduces changes of the gain and the bandwidth with the temperature.

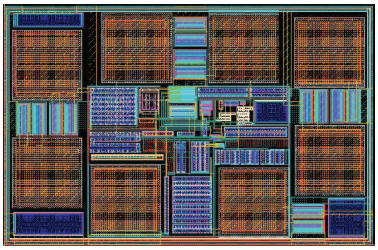


Fig4、The layout of TIA

Assuming the capacity of PD is 0.25pF, all power and bias voltage is fixed, the simulation results of TIA are that: the bode diagram is shown as fig 5(a); the gain is 82.27dB; the -3dB bandwidth is 1.8GHz; the power is about 118mW. Simulate the output noise, the wave is recorded as sout(f), and the output noise from 10MHz to 10GHz is like fig5(b). After the conversion, the maximum input referred noise is about 0.8 $pA\sqrt{Hz}$. According to the conversion relations of the input referred current noise and the output noise. The input referred current noise is calculated about 125nA.

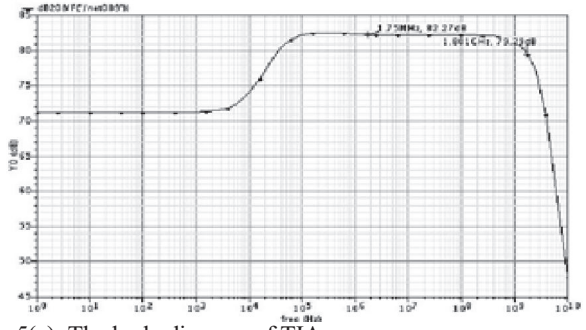


Fig. 5(a). The bode diagram of TIA

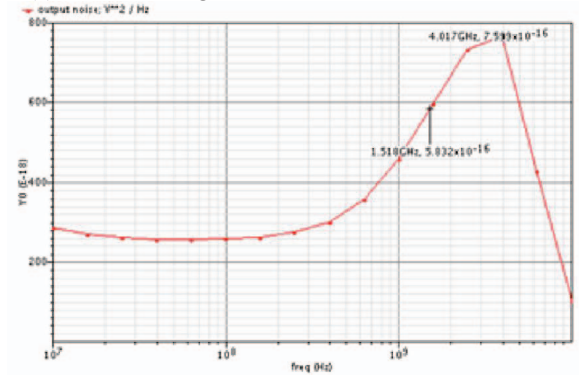


Fig. 5(b). The output noise of TIA

Fig 6 shows the eye patterns of 2.5Gbps when the photocurrent is 5uA, 10uA, 500uA and 2mA Pseudo Random Binary Sequences(PRBS) signal(The bias level of M2 changes manually). This illustrates TIA can work well when the photocurrent of PD ranges from 5uA to 2mA.

TABLE I

PERFORMANCE SUMMARY AND COMPARISON WITH OTHER WORKS

Author	Gain (dBÙ)	BW or Data Rate	PD cap. (fF)	Input-ref. noise ($pA/\sqrt{Hz}$)	Area (mm2)	Power (mW)	Technology
[15]2007	59	10Gb/s	150	25	0.41	18	0.18um CMOS
[16]2008	75	7.2GHz	450	-	1.35×0.69	91.8	0.18um CMOS
[17]2009	70.88*	1.82GHz	800	-	0.32	40	0.18um CMOS
[7]2011	80.5	0.95GHz	500	7.5	0.297	77.9	0.35um CMOS
[18]2011	28	2.488Gb/s	-	-	1×1.3	185	0.25um SiGe BiCMOS
[19]2012	61.4	2.9GHz	1000	26.8	0.18×0.18	4.9	0.18um CMOS
[20]2012	85	650MHz	-	161*	1.33	102	0.6um BiCMOS
[21]2012	68.6	10.9GHz		16.28	0.91×0.7	416	1um GaAs
this work	82.3	1.8GHz	250	0.8	0.012	118	0.13um CMOS

* are calculated from respective literature.

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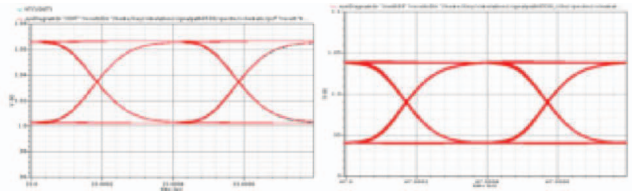


Fig. 6(a). The eye pattern of 5uA PRBS signal

Fig. 6(b). The eye pattern of 10uA PRBS signal

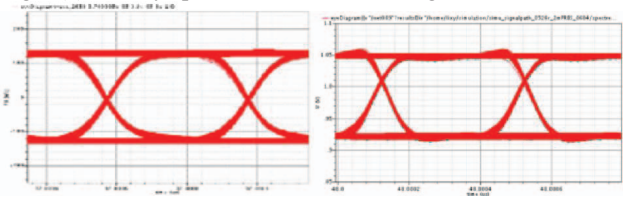


Fig. 6(c). The eye pattern of 500uA PRBS signal

Fig. 6(d). The eye pattern of 2mA PRBS signal

The performance of this TIA and other works are compared in table I.

It can be seen from table I that the area and the input referred current noise are excel others much, while its bandwidth is 1.8GHz and it owns high gain. It's suitable for low-noise optical fiber communication system. From eye patterns, they show the can be used for 2.5Gbps optical fiber communication system.

V. CONCLUSION

This paper designed a low-noise high-gain TIA with high dynamic range in 0.13ì m CMOS. It uses an individual power for the input stage to supply voltage to accommodate the level of PD, and uses a level shifting circuit to transmit the signal to the subsequent circuit. DCrestore is adopted to adjust static work points for changing PD DC current, and MOS working in the linear region is used to reduce the input signal in the input stage to realize high dynamic range. At last, key components are analyzed and optimized to reduce the input referred current noise. This TIA was validated in 0.13ì m CMOS. The simulation results showed that the maximum gain is 82.27dBÙ, the input referred current noise is 125nA in the range of 1Hz ~5GHz, the -3dB bandwidth is 1.8GHz, and it can work in 2.5Gbps communication system. The measure current ranges from 5uA to 2mA.

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